

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Mathematics
2. Subject Name: Functional Analysis L-T-P: 3-1-0 Credits: 4
3. Pre-requisites: Real Analysis
4. Syllabus and reference books:

Syllabus:

1. Normed Linear Spaces, Banach spaces, Continuity of Linear Maps
 2. Bounded linear maps and bounded linear functionals, dual spaces, reflexivity
 3. Hahn-Banach theorem, uniform boundedness principle, open mapping and closed graph theorems, weak convergence, Compact linear maps
 4. Inner product spaces, Hilbert spaces, orthonormal sets, Projection and Riesz representation theorem
 5. Bounded linear operators on Hilbert spaces- Bounded Operators and Adjoints, Normal, Unitary and Self-Adjoint Operators
- Around 12 tutorial classes will be taken

Reference Books:

1. Functional Analysis by B.V. Limaye, New Age International Publishers Limited, 2nd edition, New Delhi (1996)
2. A first course in Functional Analysis by M. T. Nair, PHI Publication (2014)
3. Introductory Functional Analysis with Applications by E. Kreyszig, Wiley India Pvt. Limited (2007)
4. A course in Functional Analysis by J. B. Conway, Springer, 2nd edition (1989)
5. Functional analysis by W. Rudin, McGraw-Hill, 2nd edition, New York (1991)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Normed Linear Spaces, Banach spaces, Continuity of Linear Maps	6
2.	Bounded linear maps and bounded linear functionals, dual spaces, reflexivity	7
3.	Hahn-Banach theorem, uniform boundedness principle, open mapping and closed graph theorems, weak convergence, Compact linear maps	15
4.	Inner product spaces, Hilbert spaces, orthonormal sets, Projection and Riesz representation theorem	6
5.	Bounded linear operators on Hilbert spaces- Bounded Operators and Adjoints, Normal, Unitary and Self-Adjoint Operators	6
Total number of hours		40